

Real-Time Object Detection Using Matched Filtering on FPGA

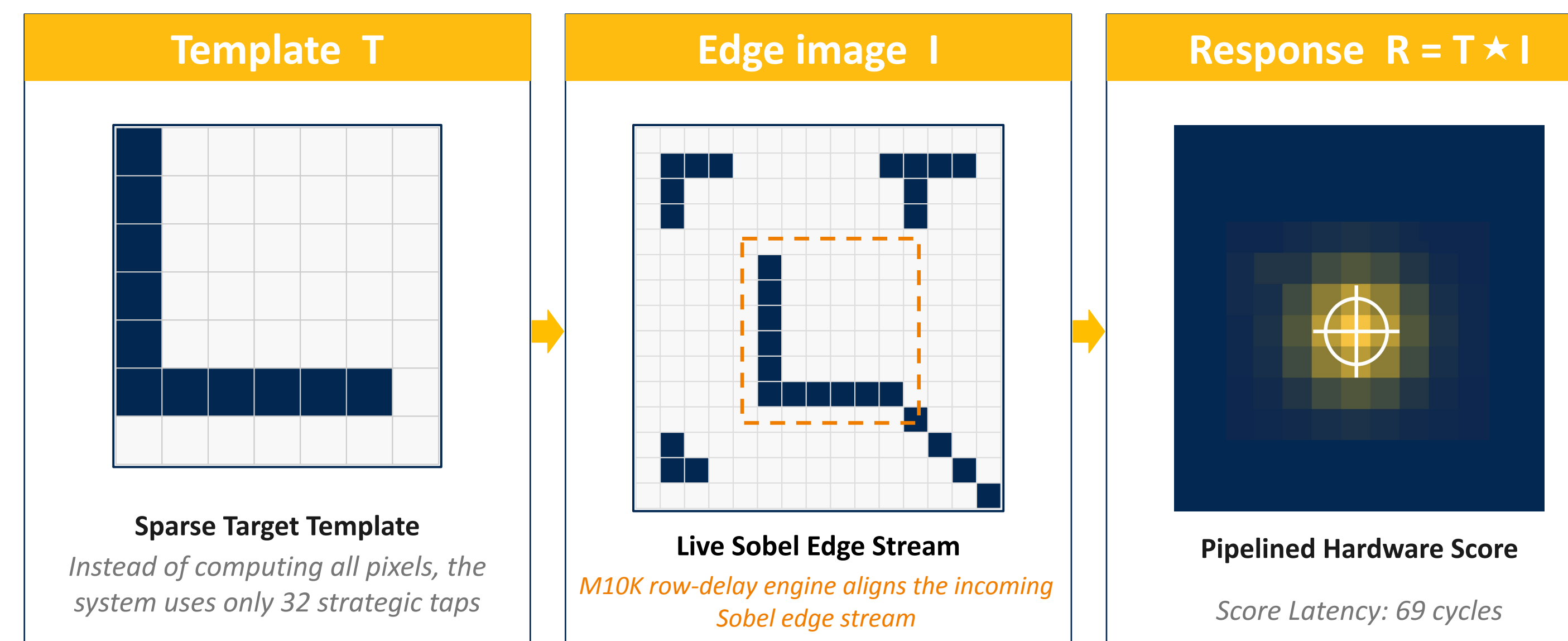
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Introduction

❖ Object detection is a fundamental task in computer vision and AI

- Traditional systems use:
 - CPUs and GPUs
- Our project uses:
 - FPGA-based matched filtering
- Benefits:
 - Low latency
 - Real-time processing

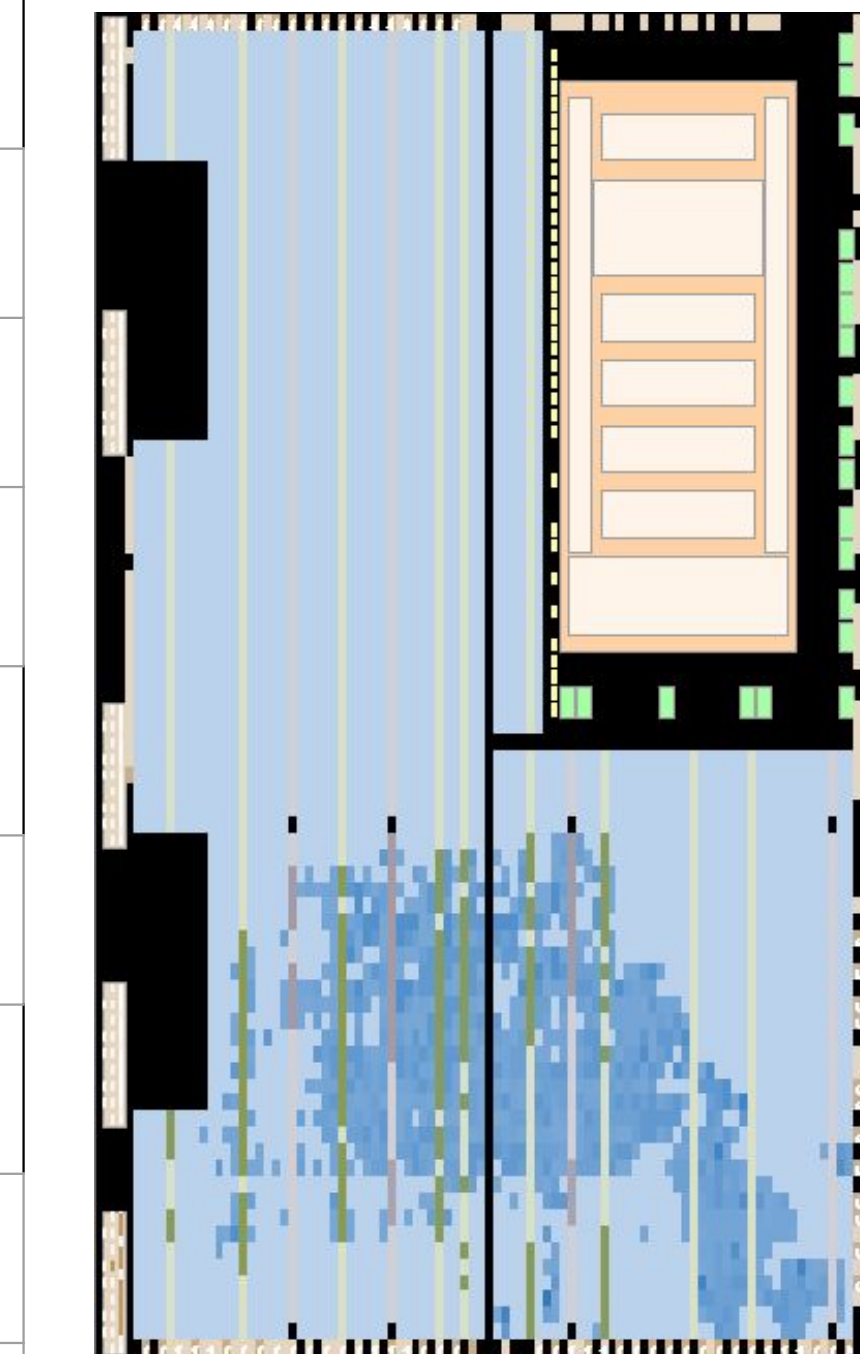
Matched Filtering



The FPGA implements a fully streaming sparse Sobel matched filter, using M10K row-delay buffers and weighted template taps to compute real-time correlation scores with 69-cycle detector latency.

Results

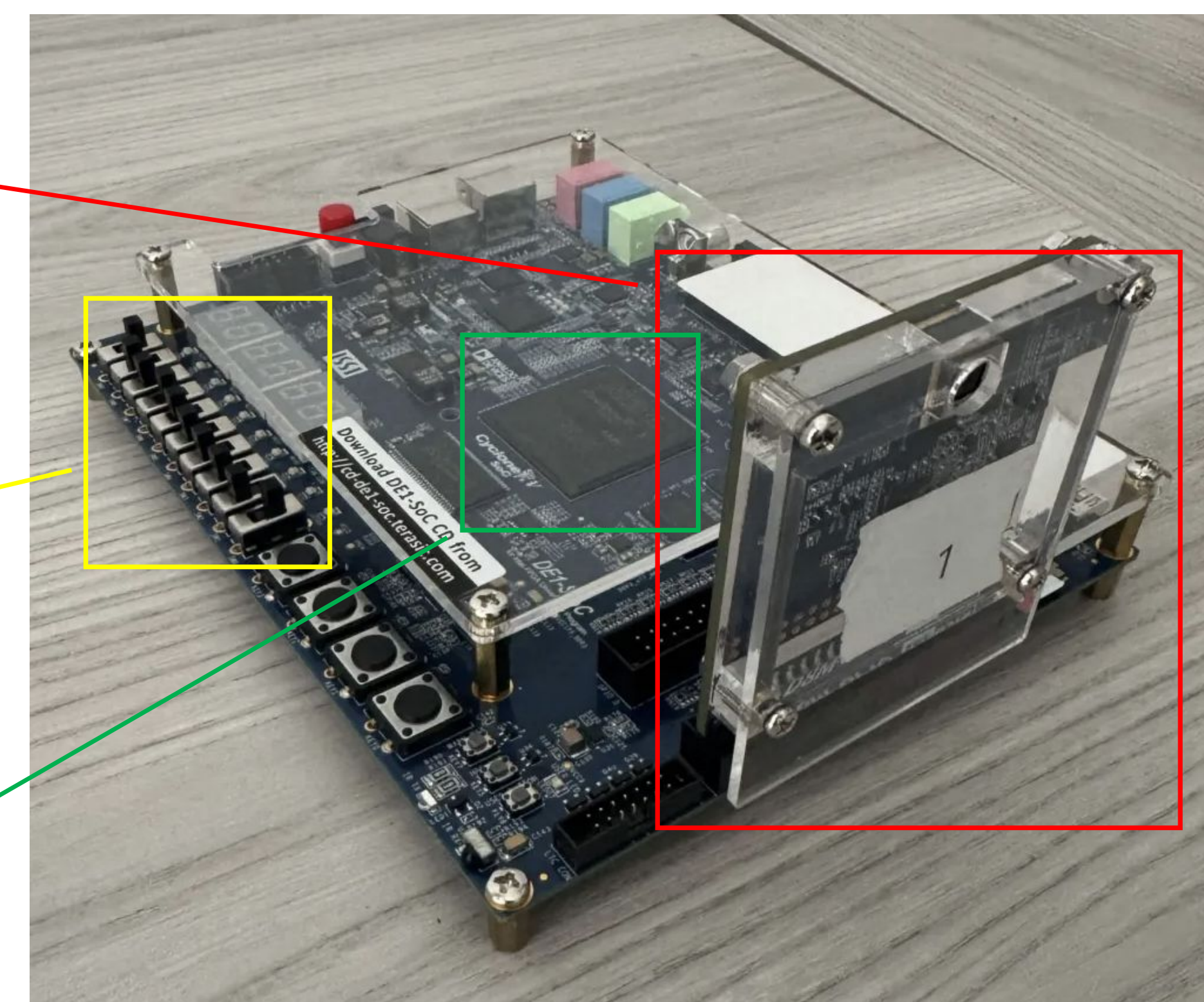
Detector Resolution	320 x 240
Template	32 x 32
Frame rate	59.5 FPS
Latency	69 cycles / 2.76 μ s
ALMs	12%
DSP Blocks	23%
M10K RAM Blocks	29%
Post-fit Fmax	47.85 MHz
Power (TDP)	524.16 mW



Quartus FPGA Floorplan

- ❖ CPU/GPU object detection works for phones and laptops
- ❖ But too slow and power-hungry for small autonomous systems
- ❖ Our FPGA achieves 59.5 FPS at 0.5 W
- ❖ Practical for automation and embedded cameras

Why FPGA?



Our hardware: Altera DE1-SoC development board with the D8M camera module mounted on the GPIO header

Camera Input

Live 50MHz RGB pixel stream captured directly from the D8M camera module

FPGA Hardware Processing

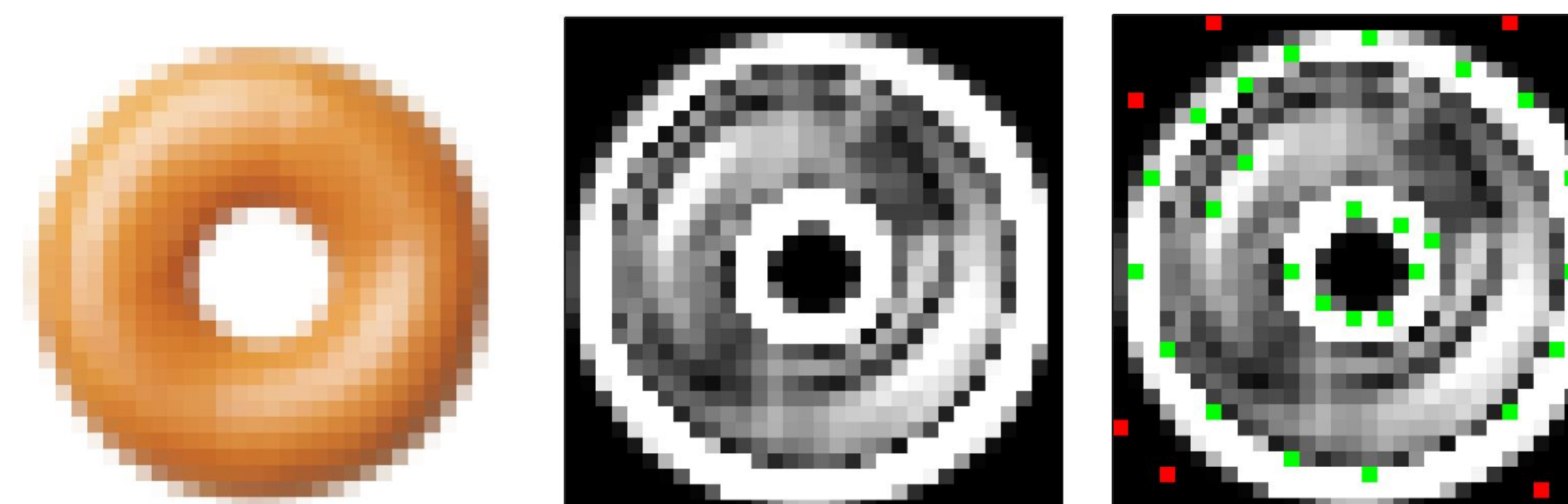
Cyclone V chip performs the full image pipeline in hardware

User Controls

Switches and push buttons select VGA display modes and adjust detection thresholds in real time

Sparse Template

- ❖ 32x32 Template reduced to strongest edge taps
- ❖ 32 signed taps: 81% positive, 19% negative
- ❖ Preserves edge shape with far fewer comparisons
- ❖ 96.9% fewer weights reduces LAB/DSP usage
- ❖ Enables fast 1-cycle streaming FPGA detection



Object Image

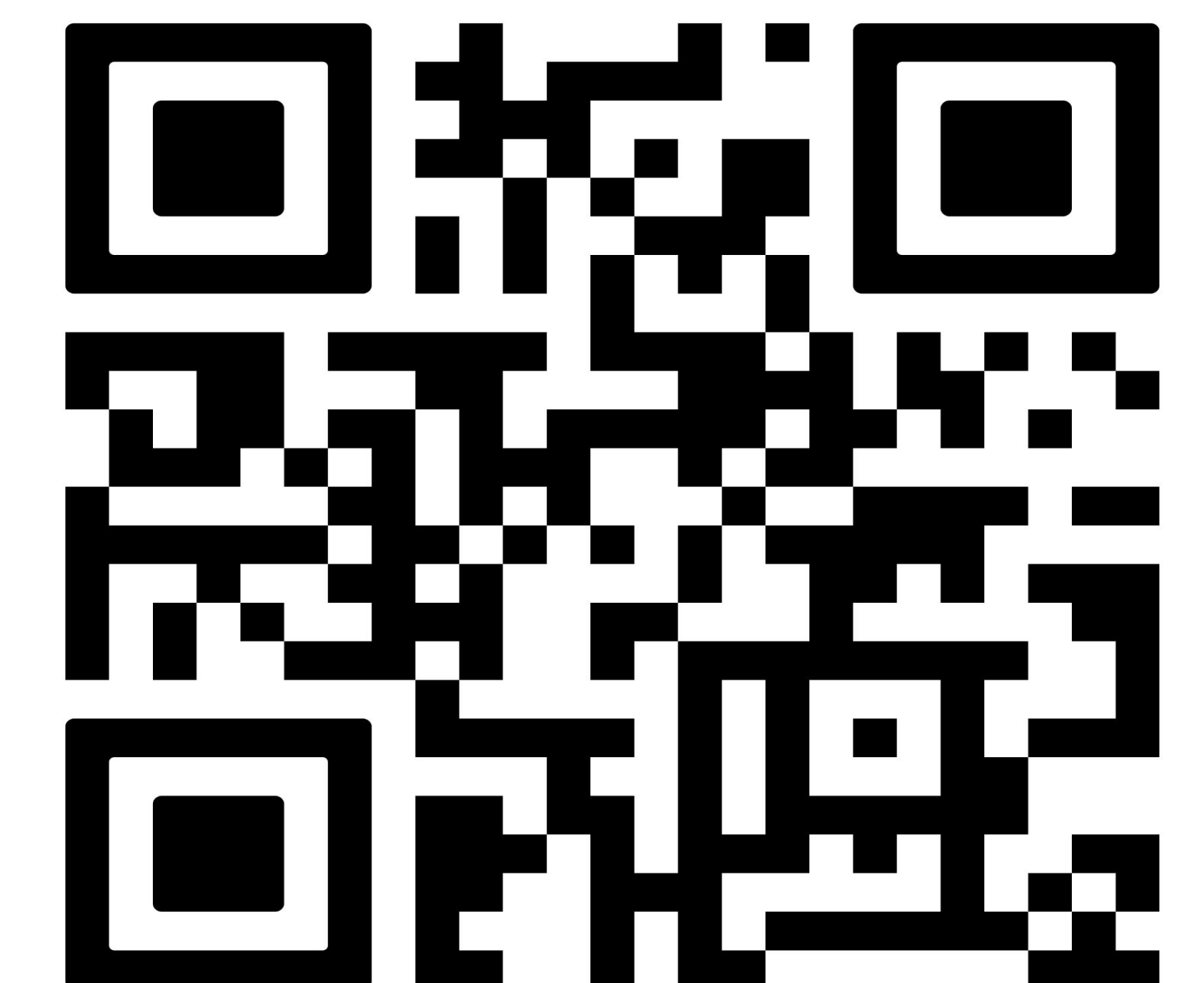
Dense Template

Sparse Template

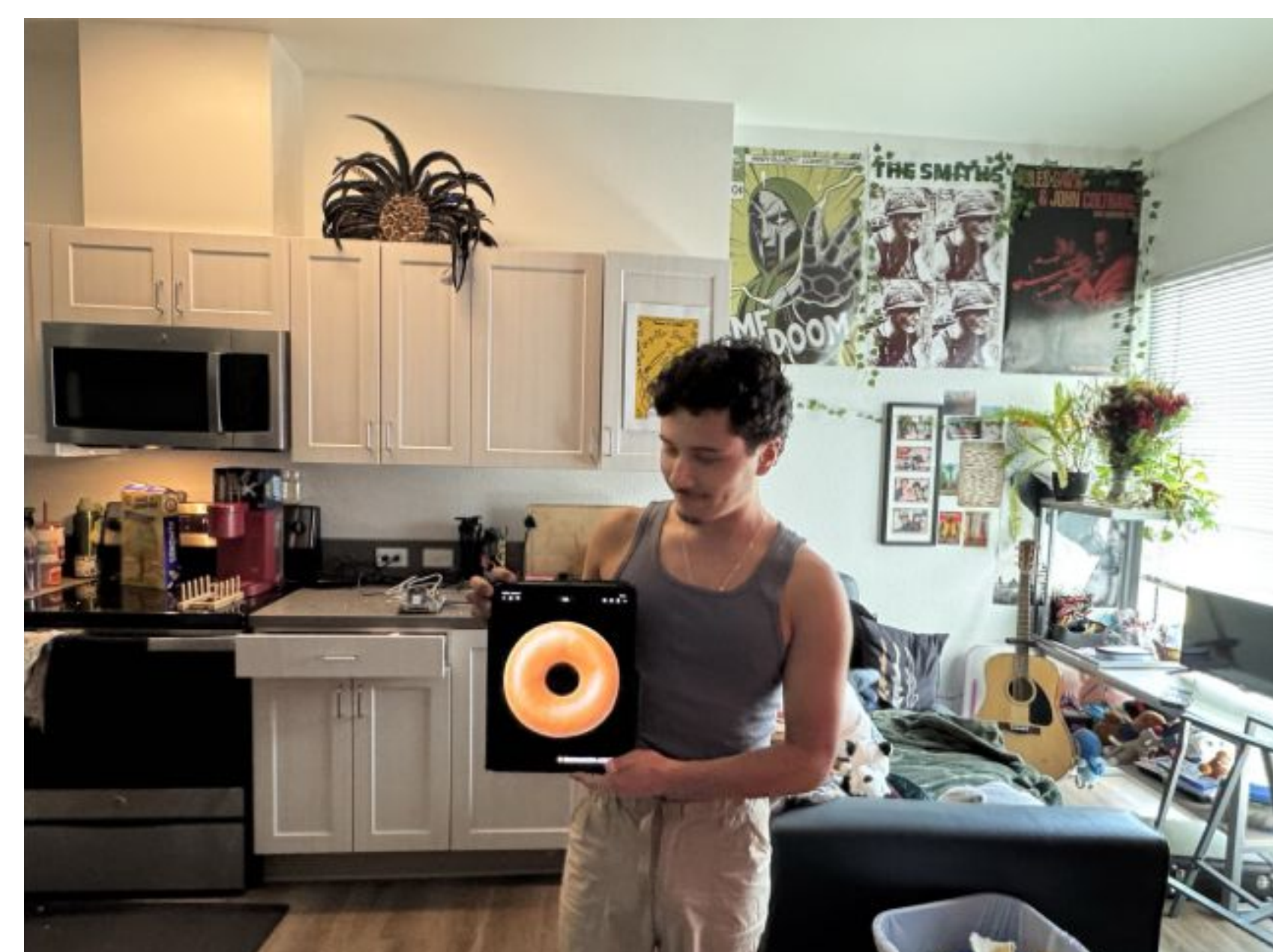
Ongoing Work

- ❖ Implement a rotational matched filter
 - Detect the object no matter the orientation
- ❖ Optimize line buffers and tap count for lower M10K/DSP usage
- ❖ Extend matcher to support multiple objects at once
- ❖ Add temporal tracking to reduce false positives
- ❖ Implement automated thresholding for changing lighting conditions

Code



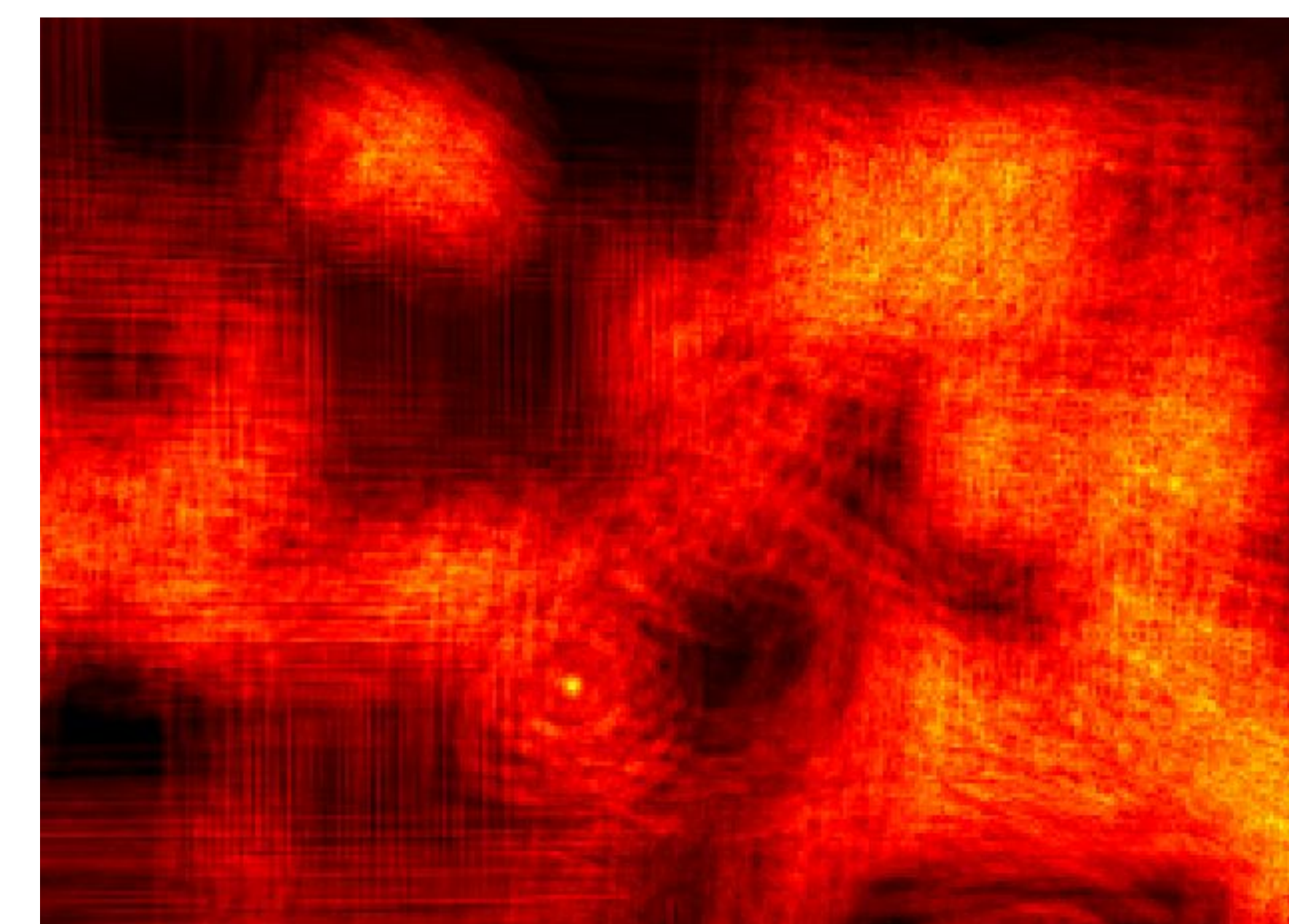
Object Tracking Pipeline



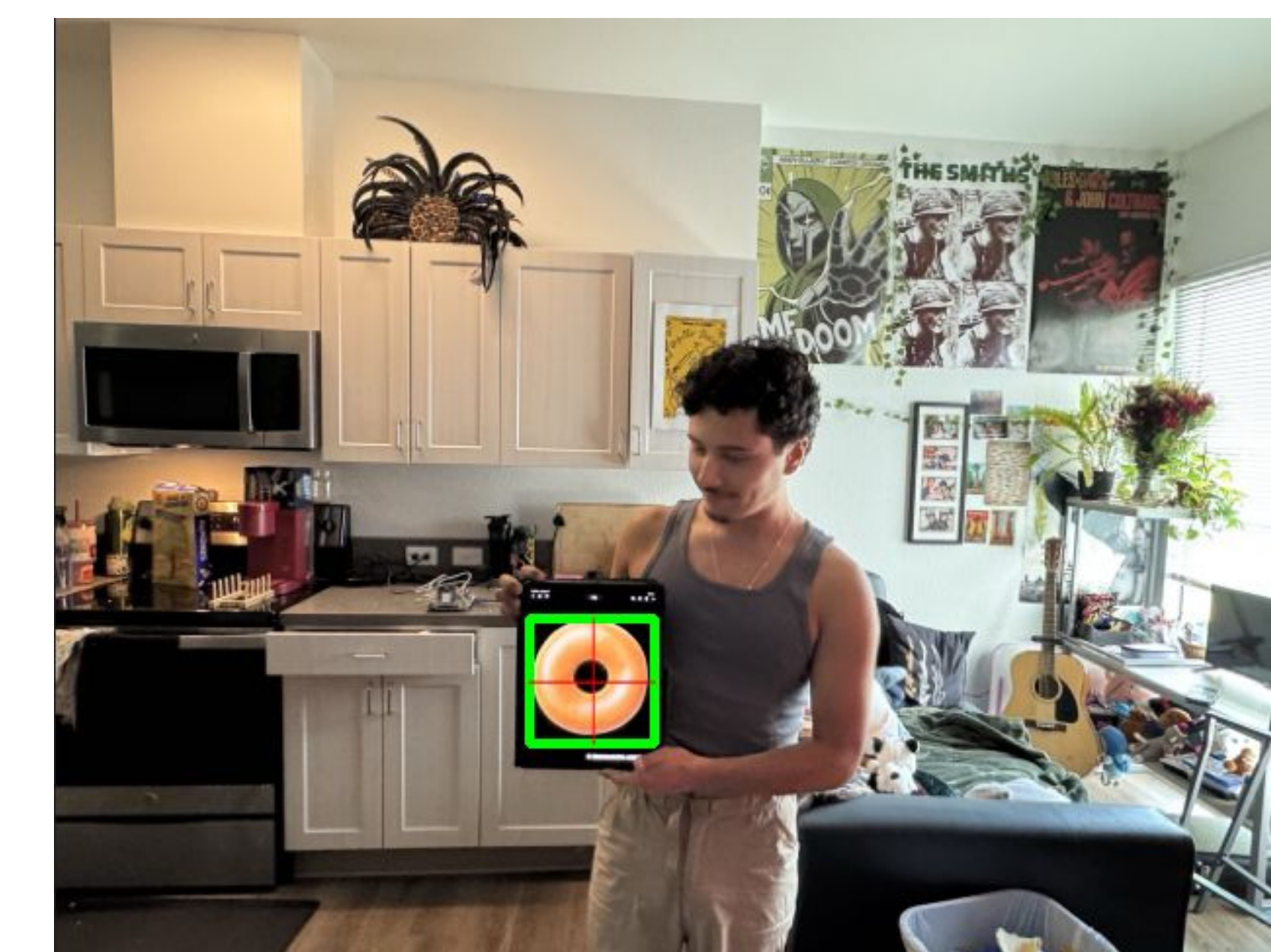
Original Camera Input
640x480 Raw RGB



Sobel Edge Filter
320x240 Sobel magnitude $|G_x| + |G_y|$



Match Score Heatmap
320x240 score RAM $\rightarrow$ 640x480 VGA | White = match



Detected Object
640x480 Raw RGB + Box overlay

References

- [1] Intel/Altera, "Altera Phase-Locked Loop (Altera PLL) IP Core User Guide," 2017. [Online]. Available: <https://www.intel.com/content/www/us/en/docs/programmable/683359/17-0.html>
- [2] Terasic Inc., "D8M-GPIO User Manual," v1.1.2, 2018. [Online]. Available: <https://www.terasic.com.tw/cgi-bin/page/archive.pl?Language=English&No=1011&PartNo=4>
- [3] Z. Dong, D. Li, and B. Baas, "EEC180/181 Tutorial: Using the D8M-GPIO 8-Megapixel Camera," UC Davis, 2025. [Online]. Available: <https://www.ece.ucdavis.edu/~bbaas/181/tutorials/camera.html>